

MASTER PROJECT BUILD BLUEPRINT

Intelligent Land Record Digitization, Validation & Urban Cadastral Mapping Platform

Document set: Unified Master Blueprint v2.0

Scope: SIH 12 + SIH 18 combined platform

Canonical rule: This document must not introduce a technology, endpoint, role, state, table, or handoff that conflicts with the Master Blueprint.

0. Purpose and reconciliation rules

This is the authoritative implementation baseline for the combined SIH 18 land-record digitization problem and SIH 12 urban cadastral mapping problem. It consolidates the previously supplied material into one internally consistent contract. If an older role PDF, diagram, endpoint list, or roadmap conflicts with this document, this version wins and the older artifact must be updated rather than interpreted differently.

Important reconciliation decisions

- Engineering-team responsibilities are separate from application RBAC roles. The six team members are implementation owners; ADMIN/OFFICER/REVIEWER/SURVEYOR/VIEWER are system-user roles.
- Person numbering is fixed: P1 Architecture/Integration; P2 Document AI; P3 Validation/Data Quality; P4 GeoAI/Cadastral; P5 Backend/Security/Deployment; P6 Frontend/Web-GIS/Verification.
- The backend is the single public integration boundary. AI/GIS services do not expose their own public security model to the browser.
- All heavy OCR/HTR, GeoAI and validation work is asynchronous and represented by job/workflow IDs.
- Image-derived parcel boundaries are preliminary until authorized GIS/survey verification; the system never invents official survey identifiers from imagery.
- Original documents/source imagery are immutable. Corrections are versioned overrides, never destructive edits.
- Every AI output carries confidence, provenance/source references, model_version and processing timestamp.

1. Combined Problem Statement

The platform addresses two connected operational problems: (1) legacy land records exist as scanned PDFs, handwritten registers, maps and historical documents that are difficult to digitize reliably; and (2) urban cadastral preparation relies heavily on manual interpretation of drone/ORI/DSM/DTM imagery and field Ground Truthing. The combined solution converts document evidence into structured land-record data and converts geospatial evidence into reviewable cadastral features, then validates and links the two streams.

2. Scope

Area	In scope	Out of scope / control
Legacy records	Scanned PDFs, images, handwritten registers, maps, cadastral records, historical documents	No destructive alteration of originals
Languages	Benchmarked printed/handwritten recognition for selected major Indian languages and mixed/degraded samples	No claim of universal language accuracy without benchmark evidence
Fields	Owner details, survey/khasra/khata, area/unit, village/tehsil/district, land classification, ownership, mutation, registration	No invented official identifiers
GeoAI	Drone imagery, ORI, DSM/DTM, existing GIS parcel layers, GT, GNSS/CORS survey-control metadata	No legal encroachment determination from imagery alone
Outputs	Structured records, preliminary parcels/buildings/roads/pathways/land-use, GeoJSON/GeoPackage/PostGIS/GeoServer layers	No publication of unverified imagery boundaries as legally authoritative
Validation	Business rules, master matching, duplicates, cross-source consistency, spatial/topology checks	No silent acceptance of high-severity conflicts
Human review	Low-confidence/conflicting/high-severity records and GIS edits	No hidden AI-only approval path
Integration	Versioned REST/OpenAPI APIs, GIS/LRMS/DILRMP integration adapters	No placeholder integration represented as a live government connection

3. Functional Requirements

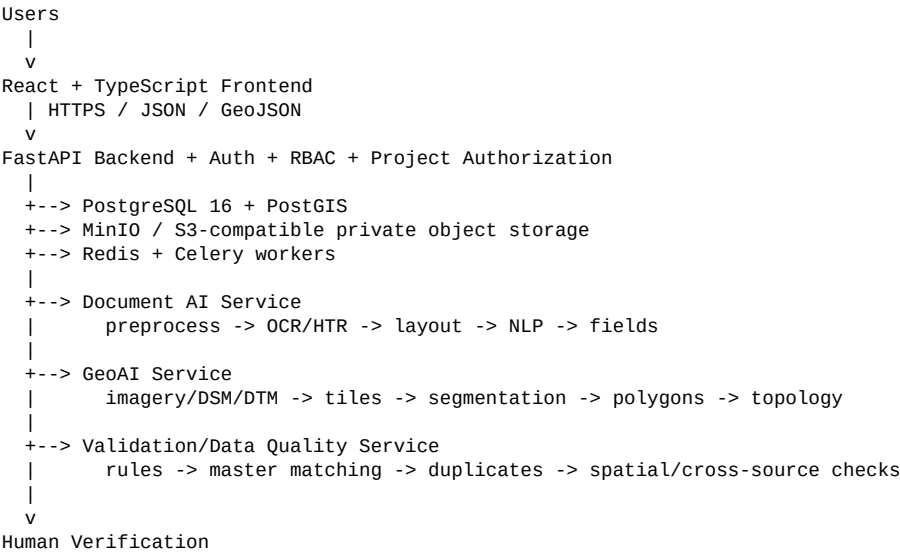
FR-01 — Secure upload/register supported documents and geospatial assets with metadata.

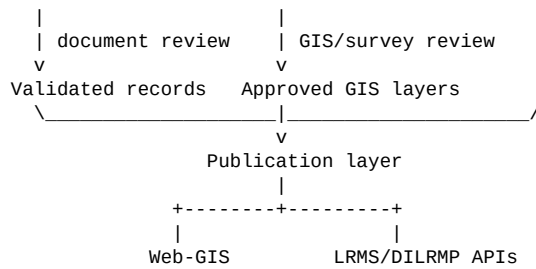
- FR-02** — Preprocess pages without changing originals.
- FR-03** — Perform benchmarked multilingual printed and handwritten OCR/HTR.
- FR-04** — Perform layout analysis and structured field extraction.
- FR-05** — Attach field-level confidence and provenance.
- FR-06** — Validate using rules, master data, duplicate detection and cross-source/spatial evidence.
- FR-07** — Provide human-assisted review and versioned correction.
- FR-08** — Ingest geospatial inputs and validate CRS/georeferencing.
- FR-09** — Extract preliminary parcels, building footprints, roads/pathways and supported land-use classes.
- FR-10** — Polygonize, repair and topology-check geometries.
- FR-11** — Compare predicted features with existing cadastral layers and create reviewable link/change flags.
- FR-12** — Expose GIS-ready outputs and Web-GIS editing/review.
- FR-13** — Enforce backend RBAC and project-scoped authorization.
- FR-14** — Maintain audit logs, model versions, immutable originals and versioned corrections.
- FR-15** — Provide dashboards, monitoring, job status, retries and recovery.
- FR-16** — Expose documented APIs and integration events.

4. Non-Functional Requirements

Category	Required behavior
Accuracy	Measure OCR/HTR, extraction, validation and GeoAI performance on representative labeled datasets.
Reliability	Long-running jobs expose status/progress/errors and support retry/recovery without duplicate results.
Security	Backend-enforced RBAC, project isolation, protected storage, safe uploads, secrets management and secure logs.
Traceability	Every AI field/issue/geometry has source evidence, model version and timestamps.
Scalability	Asynchronous workers and object storage; never block API requests with heavy AI/GIS processing.
Data integrity	Original source files immutable; corrections and model outputs versioned.
Interoperability	REST/OpenAPI, GeoJSON, PostGIS, GeoServer, GeoPackage and common raster/vector formats.
Observability	Health/readiness, metrics, structured logs, correlation IDs, job status and diagnostics.
Accessibility	Responsive UI, keyboard navigation, focus states and screen-reader labels.

5. System Architecture





6. Two authoritative workflows

6.1 Document workflow

1. Upload/register document and metadata
2. Store original immutably
3. Create asynchronous processing job
4. Preprocess pages
5. OCR / HTR
6. Layout analysis
7. NLP / structured field extraction
8. Normalization
9. Attach field confidence + provenance
10. Validation
11. Route low-confidence/conflicting/high-severity cases to review
12. Approve / correct / reject / reprocess
13. Revalidate after corrections
14. Publish
15. Audit

6.2 GeoAI workflow

1. Upload/register ORI, drone, DSM, DTM, existing GIS, GT and survey-control data
2. Validate format, CRS/EPSG, bounds, GSD/pixel size, NoData and georeferencing
3. Create spatially aware overlapping tiles
4. Run segmentation/object detection
5. Merge predictions while preserving world transforms
6. Polygonize masks
7. Controlled simplification and geometry repair
8. Calculate area in an appropriate projected CRS
9. Topology validation: validity, self-intersection, overlaps, gaps, consistency
10. Compare against existing cadastral layers and create change/link flags
11. Store draft/provisional GIS features with confidence, provenance and model version
12. GIS/survey review
13. Publish GIS-ready outputs
14. Audit

6.3 Shared state model

UPLOADED -> QUEUED -> PROCESSING -> EXTRACTED -> VALIDATING -> REVIEW_REQUIRED -> VALIDATED -> PUBLISHED
 \-> FAILED -> QUEUED (retry from failed stage)

Rules: illegal transitions are rejected; every stage writes workflow_steps; every transition creates an audit event; id

7. Six-Person Ownership Model

Owner	Responsibility	Primary handoff
Person 1	Project Lead, System Architecture & Integration	Contracts, workflow events, state machine, integration test evidence
Person 2	Document AI / OCR / HTR	OCR blocks, fields, confidence, bounding boxes, model version
Person 3	Validation / Data Quality	Validation report, issue IDs, severity, expected/observed values
Person 4	GeoAI / Urban Cadastral / GIS Processing	Layers, geometries, confidence, topology errors, model version
Person 5	Backend / Database / Security / Deployment	Stable APIs, migrations, auth/RBAC, storage, job infrastructure
Person 6	Frontend / Web-GIS / Dashboards / Human Verification	User workflows, review actions, map edits, dashboard views

8. Technology Baseline

Layer	Canonical choice	Rule
Frontend	React + TypeScript + Vite + React Router + TanStack Query	Do not access DB directly.
Backend	Python 3.12 + FastAPI + Pydantic v2 + SQLAlchemy 2 + Alembic	All public APIs under /api/v1.
Database	PostgreSQL 16 + PostGIS	Spatial indexes; transactional source of truth.
Async	Redis + Celery	OCR/GeoAI/validation jobs must be asynchronous.
Storage	MinIO / S3-compatible object storage	Private; short-lived signed URLs.
Document AI	OpenCV + benchmark-selected OCR/HTR + PyTorch + Hugging Face	Model/engine selected by benchmark.
NLP	spaCy + Hugging Face Transformers + Indic NLP	Structured extraction + normalization; preserve originals.
GeoAI	PyTorch + SegFormer/U-Net/DeepLab + YOLO/segmentation as benchmark-selected	Benchmark-selected segmentation/object detection.
Geospatial	GDAL + Rasterio + GeoPandas + Shapely	Preserve CRS/transforms; projected CRS for area.
GIS	GeoServer + OpenLayers/Leaflet; QGIS for validation	Web visualization/editing; QGIS for export validation.
Monitoring	Prometheus + Grafana + structured JSON logs	Health, readiness, metrics, logs.
Testing	pytest + Playwright + API/schema/integration tests	Unit + integration + contract + E2E + security.
Deployment	Docker Compose; CI/CD; government-approved hosting when applicable	Deployment approval depends on environment/government requirements.

9. Repository Structure

```
intelligent-land-record-platform/  
  frontend/  
    src/{api,components,pages,routes,hooks,features}  
    src/features/{auth,projects,documents,verification,parcels,gis,dashboard,audit,issues}  
    src/tests/  
  backend/  
    app/{api/v1,core,models,schemas,services,workers,repositories,audit}  
    alembic/  
    tests/  
  ai/
```

```

document_ai/{preprocessing, ocr, layout, extraction, models}
geoai/{ingestion, tiling, segmentation, polygonization, evaluation}
validation/{rules, matching, spatial, confidence}
infrastructure/
  docker/ nginx/ monitoring/
  docker-compose.yml
data/{samples, annotations}
docs/{api, architecture, deployment, team}
.env.example
README.md

```

10. Canonical Data Model

The following entities are mandatory shared concepts. Individual role documents may add implementation details, but must not rename these canonical entities without an explicit contract revision.

Entity	Purpose
users	Identity and account state
roles / permissions / user_roles / role_permissions	Application RBAC
projects / project_members	Tenant/project isolation and membership
files / documents / document_pages	Immutable source assets and page metadata
ocr_results / land_records / extractions	Document AI evidence and structured records
imagery_assets / geoai_jobs	Geospatial source registration and processing
parcels / buildings / roads / land_use_features	Draft/approved GIS features
validation_results / topology_errors	Explainable quality issues
review_tasks	Human verification work queue
model_versions	AI model provenance and release identity
processing_jobs / workflow_runs / workflow_steps	Asynchronous workflow execution
integration_events	Service handoff/event tracking
audit_logs	Immutable critical-action history
release_checklists	Deployment/release evidence

11. Canonical Field Provenance

```

field_name
original_value
normalized_value
confidence
source.document_id
source.page
source.bbox
model_version
processed_at

```

OCR confidence is not final record confidence. Record confidence combines OCR, layout, extraction, normalization and validation signals; thresholds are configurable and every review flag must state its reason.

12. Canonical Validation Rules

Rule family	Checks	Typical routing
Required fields	Required jurisdictional fields present	Review if missing
Format	Identifiers, dates and allowed formats	Review if invalid
Area	Positive area, recognized unit, normalized value	Review if invalid
Master matching	Exact match then controlled fuzzy fallback	Review if unresolved
Duplicates	Duplicate survey/khasra/khata/record identifiers	High severity if collision

Rule family	Checks	Typical routing
Identifier consistency	Document/database/GIS identifiers agree	Review on conflict
Spatial validity	Self-intersection, invalid rings, missing geometry	Block publication
Topology	Overlaps, gaps, inconsistent boundaries	GIS review
Cross-source	Recorded area vs GIS area; document vs database	Review on material mismatch
Confidence	Field/record below configured threshold	Create review task

13. Human Verification Policy

1. Reviewer opens task and confirms project scope and permissions.
2. Inspect original source and evidence bounding boxes; never rely on AI text alone.
3. Compare extracted values with document evidence and, where available, GIS evidence.
4. Review validation issues and severity.
5. Choose Approve, Correct, Reject, Reprocess, Comment or Escalate.
6. For correction, preserve before value, after value, reason, actor, timestamp and request/job ID.
7. Revalidate corrected data before publication.
8. Publish only when required validation conditions are satisfied.
9. Audit every critical action.

14. RBAC: Application Roles vs Engineering Roles

Application roles are not the same as the six-person engineering assignments. Backend authorization is authoritative; hiding a frontend button is never sufficient.

Role	Primary responsibility	Sensitive actions
ADMIN	System/project/user administration	Manage users/roles, all project actions, audit and release controls
OFFICER	Operational land-record processing	Upload/process records, correct fields, run validation, operational exports
REVIEWER	Quality and approval workflow	Review, correct, approve/reject, resolve review issues, approve GIS where authorized
SURVEYOR	Field/GIS verification	Upload survey-control/imagery, run GeoAI, edit draft geometry, inspect topology
VIEWER	Read-only approved information	View permitted approved records/layers and dashboards; no draft access

14.1 Permission matrix

Permission	ADMIN	OFFICER	REVIEWER	SURVEYOR	VIEWER
project:read	YES	YES	YES	YES	YES
project:create	YES	YES	NO	NO	NO
project:update	YES	YES	NO	NO	NO
project:member_manage	YES	NO	NO	NO	NO
document:upload	YES	YES	NO	NO	NO
document:read	YES	YES	YES	YES	YES
document:process	YES	YES	NO	NO	NO
document:reprocess	YES	YES	NO	NO	NO

Permission	ADMIN	OFFICER	REVIEWER	SURVEYOR	VIEWER
field:read	YES	YES	YES	NO	NO
field:correct	YES	YES	YES	NO	NO
record:read	YES	YES	YES	YES	YES
record:publish	YES	NO	NO	NO	NO
validation:run	YES	YES	YES	NO	NO
validation:resolve	YES	YES	YES	NO	NO
review:read	YES	YES	YES	NO	NO
review:act	YES	NO	YES	NO	NO
imagery:upload	YES	YES	NO	YES	NO
geoai:process	YES	NO	NO	YES	NO
geo:read	YES	YES	YES	YES	YES
geo:edit_draft	YES	NO	NO	YES	NO
geo:approve	YES	NO	YES	NO	NO
export:read	YES	YES	YES	YES	YES
dashboard:read	YES	YES	YES	YES	YES
audit:read	YES	YES	NO	NO	NO
user:manage	YES	NO	NO	NO	NO
role:read	YES	NO	NO	NO	NO
role:manage	YES	NO	NO	NO	NO
system:admin	YES	NO	NO	NO	NO

Security note: exact legal/administrative approval authority must be configured by the responsible authority. The matrix above is the implementation baseline, not a claim about statutory authority.

15. API Contract Rules

- All public endpoints are under /api/v1 except liveness/readiness/metrics paths defined below.
- All responses contain stable IDs and machine-readable status values.
- Long-running work returns job_id/workflow_run_id and is polled through status endpoints.
- Pagination is required for list endpoints.
- Errors use a consistent machine-readable structure with request_id.
- Private files are never public; download endpoints authorize and return short-lived signed URLs.
- Project-scoped authorization is checked server-side for every project-owned resource.

15.1 Endpoint catalogue

Auth

Method	Endpoint	Purpose
POST	/api/v1/auth/login	Authenticate and issue access token
POST	/api/v1/auth/refresh	Refresh session/token
POST	/api/v1/auth/logout	End session
GET	/api/v1/users/me	Current user, roles and project permissions

Users/Roles

Method	Endpoint	Purpose
GET	/api/v1/users	List users according to caller permission
POST	/api/v1/users	Create user
PATCH	/api/v1/users/{user_id}	Update user
GET	/api/v1/roles	List application roles/permissions

Projects

Method	Endpoint	Purpose
POST	/api/v1/projects	Create project
GET	/api/v1/projects	List projects with pagination and role filtering
GET	/api/v1/projects/{project_id}	Project metadata and permissions
PATCH	/api/v1/projects/{project_id}	Update project metadata/configuration
POST	/api/v1/projects/{project_id}/members	Add project member
DELETE	/api/v1/projects/{project_id}/members/{user_id}	Remove project member
GET	/api/v1/projects/{project_id}/summary	Project counts/backlog
POST	/api/v1/projects/{project_id}/workflows	Start document/cadastral workflow

Files/Documents

Method	Endpoint	Purpose
POST	/api/v1/files/presign	Authorize and create short-lived upload URL
POST	/api/v1/files/complete	Complete upload after checksum/content verification
GET	/api/v1/files/{file_id}/download	Authorize and return short-lived download URL
POST	/api/v1/documents	Register document
GET	/api/v1/documents	List/filter documents
GET	/api/v1/documents/{id}	Document details
POST	/api/v1/documents/{id}/process	Start document workflow
POST	/api/v1/documents/{id}/ocr	Start OCR/HTR
GET	/api/v1/documents/{id}/ocr	OCR text/blocks
POST	/api/v1/documents/{id}/extract	Start structured extraction
GET	/api/v1/documents/{id}/fields	Extracted fields
POST	/api/v1/documents/{id}/reprocess	Create new pipeline/version
GET	/api/v1/documents/{id}/pages/{page}	Page image and regions

Records/Validation/Review

Method	Endpoint	Purpose
GET	/api/v1/records/{id}	Land record
PATCH	/api/v1/records/{id}	Versioned correction
POST	/api/v1/validation/records/{id}/run	Run validation
POST	/api/v1/validation/batch	Batch validation
GET	/api/v1/validation/records/{id}	Validation report
GET	/api/v1/validation/issues	Issue queue
GET	/api/v1/validation/issues/{id}	Issue details

Method	Endpoint	Purpose
POST	/api/v1/validation/issues/{id}/resolve	Resolve issue
GET	/api/v1/review/tasks	Reviewer queue
GET	/api/v1/review/tasks/{id}	Review task
PATCH	/api/v1/review/tasks/{id}	Approve/correct/reject
POST	/api/v1/review/tasks/{id}/submit	Submit review

GeoAI/GIS

Method	Endpoint	Purpose
POST	/api/v1/imagery/upload	Register imagery/geospatial asset
GET	/api/v1/imagery/{asset_id}	Asset metadata/status
POST	/api/v1/geoai/jobs	Create GeoAI job
GET	/api/v1/geoai/jobs/{job_id}	GeoAI status
POST	/api/v1/geoai/jobs/{job_id}/cancel	Cancel queued work
GET	/api/v1/geoai/projects/{project_id}/layers	List layers
GET	/api/v1/geoai/projects/{project_id}/parcels	List draft/approved parcels
GET	/api/v1/geoai/projects/{project_id}/buildings	Building footprints
GET	/api/v1/geoai/projects/{project_id}/roads	Road/path features
GET	/api/v1/geoai/projects/{project_id}/land-use	Land-use features
GET	/api/v1/geoai/projects/{project_id}/topology-errors	Topology issues
PATCH	/api/v1/geoai/features/{feature_id}	Versioned draft geometry edit
POST	/api/v1/geoai/features/{feature_id}/validate	Validate feature
GET	/api/v1/geoai/projects/{project_id}/export?format=geojson	GeoJSON export
GET	/api/v1/geoai/projects/{project_id}/export?format=geopackage	GeoPackage export

Workflow/Audit/Monitoring

Method	Endpoint	Purpose
GET	/api/v1/workflows/{workflow_run_id}	Workflow progress and stage history
POST	/api/v1/workflows/{workflow_run_id}/retry	Retry failed stage
POST	/api/v1/workflows/{workflow_run_id}/cancel	Cancel pending work
GET	/api/v1/integrations/events	Integration event delivery failures
GET	/api/v1/dashboard/summary	Dashboard metrics
GET	/api/v1/audit/events	Audit history
GET	/api/v1/system/metrics	Operational metrics
GET	/api/v1/health	Liveness
GET	/api/v1/ready	Dependency readiness
GET	/api/v1/metrics	Prometheus-compatible metrics

15.2 Error contract

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "The uploaded file is not supported.",
    "details": [{"field": "file", "reason": "mime_not_allowed"}],
    "request_id": "uuid"
  }
}
```

16. Integration Events

Event	Producer	Required payload
document.ocr.completed	Person 2	document_id, page_ids, model_version_id, result_uri
validation.completed	Person 3	document_id, result_count, critical_error_count, review_task_ids
geoai.completed	Person 4	project_id, layer_ids, feature_counts, topology_error_count
workflow.stage.completed	Person 1/orchestrator	workflow_run_id, stage, status, timestamp
audit.event.created	Backend	actor_id, project_id, action, entity, before/after where applicable

Integration contract rule: payload names above are canonical. Teams may add optional fields but must not rename/remove required fields without a contract version change.

17. Geospatial Contract

- Inputs: high-resolution drone imagery, ORI, DSM, DTM, existing GIS parcel layers, GT datasets and GNSS/CORS-enabled survey-control metadata.
- Validate CRS/EPSG, bounds, transform, pixel size/GSD, NoData, georeferencing and file integrity before inference.
- Use overlapping spatial tiles and preserve affine transforms through inference and merge.
- Use an appropriate projected CRS for area calculations; never calculate area in latitude/longitude degrees.
- Store draft/provisional status, confidence, source asset, model version and processing parameters.
- Run geometry repair, self-intersection, overlap and gap checks; unresolved errors become topology_errors.
- Export GeoJSON and GeoPackage; verify exported files in QGIS.
- Draft edits must not overwrite original AI predictions.

18. Document AI Contract

1. Preserve original PDF/image and register immutable source.
2. Render pages when needed; detect orientation, deskew, denoise, enhance contrast and detect/crop document regions.
3. Benchmark OCR/HTR on clean, degraded, handwritten, tabular and mixed-language samples.
4. Retain OCR text plus bounding boxes and model version.
5. Run layout analysis for tables, field-value regions, stamps, signatures and sections.
6. Extract structured fields using NLP/ML plus deterministic normalization.
7. Attach field confidence and provenance.
8. Do not treat OCR confidence as final record confidence.
9. Emit the canonical completion event.

19. Audit and Data Governance

- Record actor, timestamp, action, entity, before/after values, reason and request/job ID.
- Preserve original documents and source imagery.
- Version corrections and model outputs.
- Do not log passwords, tokens or sensitive document contents unnecessarily.
- Keep secrets in environment/secret management.
- Apply retention, backup and archival policies approved by the responsible authority.
- Never delete historical audit evidence through ordinary UI actions.

20. Deployment Architecture

```
docker-compose services:
  frontend
  backend
  worker
  postgres-postgis
  redis
  minio
  geoserver
  optional: prometheus
  optional: grafana
```

External integration adapters:
LRMS / DILRMP / GIS / government identity provider
(enabled only when credentials and approvals exist)

21. Environment Contract

```
APP_ENV=development
DATABASE_URL=postgresql+psycopg://<user>:<password>@postgres-postgis:5432/landrecords
REDIS_URL=redis://redis:6379/0
S3_ENDPOINT=http://minio:9000
S3_ACCESS_KEY=<secret>
S3_SECRET_KEY=<secret>
S3_BUCKET=land-records
JWT_SECRET=<secret-or-approved-identity-config>
OCR_MODEL_PATH=<configured-model>
GEOAI_MODEL_PATH=<configured-model>
GEOSERVER_URL=http://geoserver:8080/geoserver
```

22. Testing and Acceptance Matrix

Area	Minimum acceptance evidence
Document AI	Clean OCR, degraded image, handwritten sample, multilingual sample, confidence and provenance
GeoAI	CRS validation, tile transform, polygon validity, area, GeoJSON/GeoPackage
Validation	Required fields, duplicate, area mismatch, geometry issue, severity routing
Backend	Auth, RBAC, CRUD, jobs, retries, audit, schema validation
Frontend	Login, upload, processing, review, map, permissions, error states
Security	Viewer cannot access drafts; cross-project access blocked; signed URLs expire; invalid MIME rejected
System	Upload-to-publish, failed-job recovery, versioned correction, audit trail
Deployment	Fresh machine starts with Docker Compose; migrations/seed/reset documented

23. Implementation Roadmap

Phase	Deliverables	Owner(s)
A. Contracts	Freeze schemas, statuses, API, events, GeoJSON contract	P1 + all; P5/P4 lead implementation contracts
B. Infrastructure	Docker Compose, DB/PostGIS, Redis, MinIO, environments	P5
C. Backend foundation	Auth, RBAC, CRUD, storage, jobs, migrations, OpenAPI	P5
D. Document AI MVP	Preprocess, OCR/HTR, layout, extraction, confidence, provenance	P2
E. GeoAI MVP	Raster ingestion, segmentation, polygonization, topology, GIS exports	P4
F. Validation MVP	Rules, matching, duplicates, cross-source/spatial checks, issue schema	P3
G. Frontend/Review	Login, upload, document viewer, review workspace, Web-GIS, dashboards	P6
H. Integration	Connect all services through canonical contracts/events	P1 + all
I. Test/Security	Unit, contract, integration, E2E, RBAC, recovery, audit	P5 + P6 + all

Phase	Deliverables	Owner(s)
J. Demo/Release	Success + discrepancy + GIS + review + dashboard + audit scenario	All; P1 release owner

24. Definition of Done

- Fresh machine starts the complete stack with Docker Compose.
- Scanned land record is uploaded and processed.
- Printed, handwritten and at least one supported Indian-language sample are demonstrated.
- Structured fields and provenance are stored.
- At least one georeferenced ORI/drone sample is processed.
- Preliminary parcel geometries are stored in PostGIS and displayed in Web-GIS.
- At least one validation discrepancy and one duplicate case are demonstrated.
- Low-confidence/conflicting cases create review tasks.
- Reviewers can approve, correct, reject and reprocess.
- GIS draft edits are versioned and do not overwrite original predictions.
- Corrections and critical actions are audited.
- Dashboard metrics come from backend data.
- RBAC is enforced server-side and project isolation is tested.
- OpenAPI documentation is available.
- CI pipeline is green and E2E upload-to-publish passes.
- Recovery/seed/reset procedure is documented.

25. Master Build Instructions for an AI Coding Agent

Use the following as the controlling build prompt. It is intentionally explicit so that six agents can work in parallel without drifting from the same contracts.

You are implementing the Intelligent Land Record Digitization, Validation & Urban Cadastral Mapping Platform.

AUTHORITATIVE RULES

1. Treat this Master Blueprint v2.0 as the single source of truth.
2. Do not invent endpoints, entity names, statuses, roles, permissions, or event payloads that conflict with this document.
3. Separate engineering ownership (P1-P6) from application RBAC roles.
4. Finalize Pydantic schemas and OpenAPI contracts before dependent implementation.
5. Every AI output contains confidence, provenance/source reference, model_version and timestamp.
6. Originals are immutable; corrections are versioned.
7. Heavy processing is asynchronous and has job_id/workflow_run_id, status, progress, errors and retry.
8. Backend is the only browser-facing integration boundary; frontend never connects directly to PostgreSQL.
9. Enforce project-scoped authorization server-side.
10. Never invent official survey identifiers or publish unverified imagery boundaries as authoritative.
11. Use benchmark-driven model selection.
12. Use canonical API and event names exactly.
13. Run unit, schema, integration, security and E2E tests.
14. Do not use fake success responses. If a model/integration is unavailable, implement a clearly marked adapter/mock wrapper.
15. At each phase report changed files, run commands, test results, limitations and next handoff.
16. Before final release, execute the Definition of Done and archive evidence.

26. Critical Engineering Warnings

- Do not build frontend pages against guessed API responses.
- Do not store only a giant OCR text blob; preserve structured fields and coordinates.
- Do not overwrite original files or previous extraction/model outputs.
- Do not treat OCR confidence as final record confidence.
- Do not hard-code one document layout.
- Do not run heavy OCR/GeoAI synchronously inside API requests.
- Do not calculate area in geographic latitude/longitude degrees.
- Do not merge raster tiles without preserving geospatial transforms.
- Do not publish unverified imagery-derived boundaries as legally authoritative parcels.
- Do not silently retrain production models from reviewer corrections; corrections become controlled training/evaluation data.
- Do not bypass backend authorization because a frontend control is hidden.
- Do not present placeholder government integrations as connected production systems.